2313666 殷家兴 第七次自控作业

7-9

解：(a) 令 $G_1(s)=\frac{2}{s+2}$ $\quad G_2(s)=\frac{5}{s+5}$

$$G(s)=G_1(s)\,G_2(s)$$

$$C(s)=R(s)G(s)=R(s)\,G_1(s)\,G_2(s)$$

对 $C(s)$ 离散化，有 $\cancel{C(z)=R(z)\,G_1(z)\,G_2(z)}$ $C^*(s)=R^*(s)\,G_1^*(s)\,G_2^*(s)$

作Z变换，& $C(z)=R(z)G_1(z)G_2(z)$

$$G(z)=\frac{C(z)}{R(z)}=G_1(z)\,G_2(z)$$

$$\cancel{=2\times\frac{z}{z-e^{-2T}}+5\times\frac{z}{z-e^{-5T}}}$$

$$\cancel{=\frac{7z^2-5ze^{-2T}-2ze^{-5T}}{(z-e^{-2T})(z-e^{-5T})}}$$

$$=\frac{2z}{z-e^{-2T}}\times\frac{5z}{z-e^{-5T}}=\frac{10z^2}{(z-e^{-2T})(z-e^{-5T})}$$

(b) 令 $G_1(s)=\frac{2}{s+2}$ $\quad G_2(s)=\frac{5}{s+5}$

$$G(s)=G_1(s)\,G_2(s)$$

$$C(s)=R(s)\,G_1(s)\,G_2(s)$$

对 $C(s)$ 离散化，有 $\cancel{C(z)=R(z)\,G_1}$ $C^*(s)=R^*(s)\,G_1G_2^*(s)$

作Z变换 $C(z)=R(z)\,G_1G_2(z)$

$$G(z)=\frac{C(z)}{R(z)}=Z\left[\frac{10}{(s+2)(s+5)}\right]$$

$$=Z\left[\frac{10}{3}\times\left(\frac{1}{s+2}-\frac{1}{s+5}\right)\right]$$

$$=\frac{10}{3}\times\left(\frac{z}{z-e^{-2T}}-\frac{z}{z-e^{-5T}}\right)$$

$$=\frac{10z(e^{-2T}-e^{-5T})}{3(z-e^{-2T})(z-e^{-5T})}$$

7-10

解：(a)

$$C(s)=[E_1(s)-E_2(s)]\,G_1(s)\ ①$$

$$E_1(s)=\cancel{C(s}\ R(s)-C(s)G_3(s)\ \cancel{=R(s)-E_1(s)G_1(s)G_3(s)+E_2(s)G_1(s)G_3(s)}\ ②$$

$$E_2(s)=G_2(s)\,C(s)=E_1(s)G_1(s)G_2(s)-E_2(s)G_1(s)G_2(s)$$

分别对上式作离散化，Z变换

$$C(z)=[E_1(z)-E_2(z)]\,G_1(z)$$

$$E_1(z)=R(z)-C(z)\,G_3(z)$$

$$E_2(z)=C\,G_2(z)$$

联立上式，解得：

$$C(s)=\frac{G_1(s)R(s)}{G_1(s)G_3(s)+G_1(s)G_2(s)}$$

对其进行离散化，Z变换

$$C(z)=\frac{R(z)\,G_1(z)}{G_1(z)G_3(z)+G_1(z)G_2(z)}$$

$$E_2(s)=\frac{G_1(s)\,G_2(s)}{1+G_1(s)\,G_2(s)}E_1(s)\ ③$$

对①、②、③离散化，作Z变换

$$\begin{cases}C(z)=[E_1(z)-E_2(z)]\,G_1(z)\\ E_1(z)=R(z)-\cancel{G_1(z)E_1G_3(z)+E_2(z)G_1(z)G_3(z)}\ C(z)G_3(z)\\ E_2(z)=\frac{G_1G_2(z)}{1+G_1G_2(z)}E_1(z)\end{cases}$$

联立得：

$$\cancel{G(z)}\ \Phi(z)=\frac{G_1(z)}{1+G_2G_1(z)+G_1(z)G_3(z)}$$

(b)

$$\begin{cases} E(s)=R(s)G_1(s)-C(s) \\ B(s)=G_h(s)G_3(s)E(s) \\ C(s)=[R(s)G_2(s)+B(s)]G_4(s) \end{cases}$$

$$=R(s)G_2(s)G_4(s)+B(s)G_4(s)$$

~~对上式作离散化，有~~ $=R(s)G_2(s)G_4(s)+G_h(s)G_3(s)G_4(s)E(s)$

~~$E^*(s)=RG_1^*(s)-C^*(s)$~~

~~B^*~~ 对上式作离散化

~~联立得 $C(s)=$~~

$$\begin{cases} E^*(s)=RG_1^*(s)-C^*(s) \\ B^*(s)=G_hG_3^*(s)E^*(s) \\ C^*(s)=RG_2G_4^*(s)+E^*(s)G_hG_3G_4^*(s) \end{cases}$$

联立，得

$$C^*(s)=\frac{RG_2G_4^*(s)+RG_1^*(s)G_hG_3G_4^*(s)}{1+G_hG_3G_4^*(s)}$$

~~作Z~~ 作输出信号的采样拉氏变换

$$C(z)=\frac{RG_2G_4(z)+RG_1(z)G_hG_3G_4(z)}{1+G_hG_3G_4(z)}$$

(c)

$$B_2(z)=R(z)D_2(z)+\cancel{E(z)}\,[\cancel{E(z)+R(z)}]\,[R(z)-E(z)]D_1(z) \quad ①$$

$$B_1(s)=B_2(s)G_h(s)G_1(s) \quad ②$$

$$C(s)=[N(s)+B_1(s)]G_2(s)=N(s)G_2(s)+B_2(s)G_h(s)G_1(s)G_2(s) \quad ③$$

对②、③、~~代进~~④离散化 $E(s)=C(s)$ ④

$$B_1^*(s)=B_2^*(s)G_hG_1^*(s)$$

$$C^*(s)=NG_2^*(s)+B_2^*(s)G_hG_1G_2^*(s) \quad ⑤$$

$$E^*(s)=C^*(s) \quad ⑥$$

对⑤、⑥作输出信号的采样拉氏变换

~~$E(z)=$~~ $C(z)=NG_2(z)+B_2(z)G_hG_1G_2(z)$ ⑦

$E(z)=C(z)$ ⑧

联立①、⑦、⑧，得：

$$C(z)=\frac{NG_2(z)+R(z)\,\cancel{D_2(z)}\,[D_1(z)+D_2(z)]\,G_hG_1G_2(z)}{1+D_1(z)G_hG_1G_2(z)}$$

7-11

解：$C(z) = G(z)R(z) = \dfrac{0.53 + 0.1z^{-1}}{1 - 0.37z^{-1}} \times \dfrac{z}{z-1}$

$$= \frac{z(0.53 + 0.1z^{-1})}{(1-0.37z^{-1})(z-1)}$$

$$= \frac{z(0.53z + 0.1)}{(z-0.37)(z-1)}$$

$$c(nT) = \mathrm{Res}\left[C(z)\cdot z^{n-1}\right]_{z\to 0.37} + \mathrm{Res}\left[C(z)z^{n-1}\right]_{z\to 1}$$

$$= \lim_{z\to 0.37} \frac{z^n(0.53z+0.1)}{z-1} + \lim_{z\to 1} \frac{z^n(0.53z+0.1)}{z-0.37}$$

$$= -0.47 \times (0.37)^n + 1$$

用幂级数法进行验证，有

$$C(z) = \frac{z(0.53z+0.1)}{\cancel{0.3}(z-0.37)(z-1)} = \frac{0.53z^2+0.1z}{z^2-1.37z+1.37}$$

$$= 0.53 + 0.8261z^{-1} + 0.935z^{-2} + \cdots$$

其结果与反演法一致

7-12

解：(1) $G(s) = \dfrac{1}{s(s+5)} = \dfrac{1}{4}\left[\dfrac{1}{s} - \dfrac{1}{s+5}\right]$

作Z变换

$$G(z) = \frac{1}{4}\left(\frac{z}{z-1} - \frac{z}{z-e^{-5T}}\right) = \frac{4.0067z^2 + 0.9598z}{25z^3 - 50.1675z^2 + 25.335z - 0.1675}$$

$R(z) = \dfrac{z}{z-1}$ $\quad C(z) = G(z)R(z) = \dfrac{z^2}{4(z-1)}\left(\dfrac{1}{z-1} - \dfrac{1}{z-e^{-5T}}\right)$

$$= \frac{z^2(1-e^{-5T})}{4(z-1)^2(z-e^{-5T})}$$

$$\Phi(z) = \frac{G(z)}{1+G(z)} = \frac{0.1603z^2 + 0.0384z}{z^3 - 1.8464z^2 + 1.0518z - 0.0067}$$

$$C(z) = \Phi(z)R(z) = \frac{0.1603z^3 + 0.0384z^2}{z^4 - 2.8464z^3 + 2.8982z^2 - 1.0585z + 0.0067}$$

(2) $C(z) = \dfrac{z(1-e^{-5})}{1+z(1-e^{-5})} \cdot \dfrac{z}{z-1} = \dfrac{\cancel{z^2(1-e^{-5})}}{z^2(\cancel{1-e^{-5}) + z \cdot e^{-5} - 1}} \quad \dfrac{z^2(1-e^{-5})}{[z(1-e^{-5})+1](z-1)}$

$c(nT) =$

$\cancel{c^*(t)} = c(n) = \mathrm{Res}\left[z^{n-1}C(z)\right]_{z\to 1} + \mathrm{Res}\left[z^{n-1}C(z)\right]_{z\to \frac{1}{e^{-5}-1}}$

$$= \frac{1-e^{-5}}{2-e^{-5}} \cancel{*} - \frac{(1-e^{-5})^2}{2-e^{-5}}\left(\frac{1}{e^{-5}-1}\right)^n = \frac{1-e^{-5}}{2-e^{-5}}\left[1 + \left(\frac{1}{e^{-5}-1}\right)^{n-1}\right]$$

$$c^*(t) = \sum_{n=0}^{\infty} c(nT)\delta(t-nT) = \cancel{\sum_{n=0}^{\infty} c(n)\delta(t-n)} \sum_{n=0}^{\infty} \frac{1-e^{-5}}{2-e^{-5}}\left[1+\left(\frac{1}{e^{-5}-1}\right)^{n-1}\right]\delta(t-nT)$$

将 $C(z)$ 展开为幂级数

$C(z) = 0.16z^{-1} + 0.49z^{-2} + 0.94z^{-3} + \cdots$

对其作Z反变换

$c^*(t) = 0.16\delta(t-T) + 0.49\delta(t-2T) + 0.94\delta(t-3T) + \cdots$

(3) $\cancel{c(+\infty)}$ $c(\infty) = \lim_{n \to \infty} c(nT) = \lim_{z \to 1} (z-1)C(z)$

$$= \lim_{z \to 1} \frac{z^2(1-e^{-5})}{z(1-e^{-5})+1}$$

$$= \frac{1-e^{-5}}{2-e^{-5}}$$

先判断稳定性

闭环系统特征方程

$$z^3 - 1.8464z^2 + 1.0518z - 0.0067 = 0$$

$$z_{1,2} = 0.9167 \pm 0.4331j$$

$$z_3 = 0.0066$$

$|z_1| = |z_2| = 1.0138 > 1$ 故不稳定，无法求终值